

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Electronics and Telecommunication Engineering Summer Term Examination Academic Year: 2024-25	
Class: SY B.Tech	Subject: Data Structures	Subject Code: ESC284
Date: 19.06.2025	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- This question papers consist of 2 pages.*
- Assume suitable data whenever necessary and state it clearly.*
- Mobile phones and programmable calculators are strictly prohibited.*
- Exchange/Sharing of stationery, calculator etc. not allowed.*
- Figures to right indicate full marks.*
- The symbols carry usual meaning.*

Q.N.		Marks
Q.1	What is data structure? Why we need data structure? What are the different operations performed on data structure? What are the different types of data structures?	10
Q.2	What is stack? Why we need it? Show how to implement a stack using two queues? Analyse the run time of the stack operations.	10
Q.3	What is queue? Why we need it? Show how to implement a queue using two stacks? Analyse the run time of the queue operations.	10
Q.4	What is object oriented programming? Why we need it? Explain various elements of object oriented programming.	10
Q.5	What is binary search tree? Draw binary search tree for the following data 10, 08, 15, 12, 13, 07, 09, 17, 20, 18, 04, 05 Also show a postorder, preorder and inorder traversal of tree.	10
Q.6	What is binary tree? Construct a binary tree from the given traversal Preorder: A B D G C E H I F Inorder: D G B A H E I C F	10

Q.7	What is circular queue? Why we need it? Write a C++ code to perform various operations on it?	10
Q.8	a) Suppose there is a function with the following prototype: <pre>void f(int = 10, int = 20, int = 30, int = 40)</pre> If this function is called by passing 2 arguments to it, how can we make sure that these arguments are treated as first and third, whereas, the second and the fourth are taken as defaults. b) Create four integers, four pointers to these integers and four references to them. Store these pointers and references in two arrays and print out the values of four integers using these arrays.	10
Q.9	How to design a movie ticket booking system? Explain it with proper algorithm and C ++ code.	10
Q.10	Shanaya is given the number of visitors at her local theme park on N consecutive days. The number of visitors on i^{th} day is V_i. A day is <i>record breaking</i> if it satisfies both of the following conditions: • The number of visitors on the day is strictly larger than the number of visitors on each of the previous days. • Either it is the last day, or number of visitors on the day is strictly larger than the number of visitors on the following day. Note than the very first day could be a record breaking day! Please help shanaya to find out the number of record breaking days by writing proper algorithm and C++ code.	10

=== ALL THE BEST ===



S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.
Electronics and Telecommunication Engineering
Summer Term Examination AY 2024-25

Date: 24/06/2025
Time: 14:30-17:30

Subject: Analog Circuits and Design
Course Code: PCC-EC205

Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. All questions are compulsory.
- iii. Mobile phones and programmable calculators are strictly prohibited.
- iv. Exchange/Sharing of stationery, calculator etc. not allowed.
- v. Figures to right indicate full marks.

Q. No.	Questions	Marks	CO	BT
Q.1	Write a note on heat sinks.	10M	CO5	BT2
Q.2	Sketch and explain high frequency hybrid – π model for BJT.	10M	CO4	BT5
Q.3	Derive the expression for series-shunt feedback amplifier for A_v , A_i , R_i , R_o .	10M	CO3	BT3
Q.4	Sketch and explain CE amplifier circuit for frequency response.	10M	CO2	BT4
Q.5	Derive expression for f_β ? Calculate the bandwidth f_β and capacitance C_π of a bipolar transistor. Consider a bipolar transistor that has parameters $f_T = 20$ GHz at $I_C = 1$ mA, $\beta_o = 120$, and $C_\mu = 0.08$ pF.	10M	CO4	BT2
Q.6	What are the merits and demerits of negative feedback amplifier.	10M	CO1	BT1
Q.7	Sketch the circuit for Hartley oscillator and explain its working.	10M	CO2	BT2
Q.8	Draw the series-series configuration feedback topology and derive the R_i and R_o with and without feedback.	10M	CO4	BT5
Q.9	What are the different ideal feedback topologies.	10M	CO3	BT1
Q.10	Write a note on general frequency response characteristics of amplifiers.	10M	CO6	BT2

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO 1	Understand the fundamentals and areas of applications for the integrated circuits.
CO 2	Analyze important types of integrated circuits
CO 3	Demonstrate the ability to design practical circuits that perform the desired operations
CO 4	Understand the functioning of OP-AMP and design OP-AMP based circuits
CO 5	Understand the differences between theoretical, practical & simulated results in integrated circuits.
CO 6	Select the appropriate integrated circuit modules to build a given application.

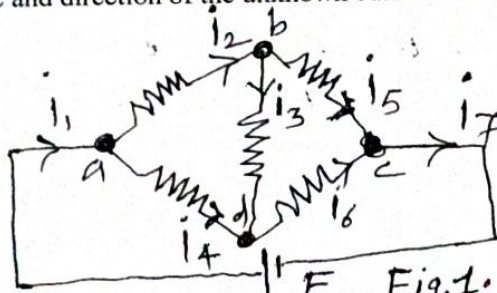


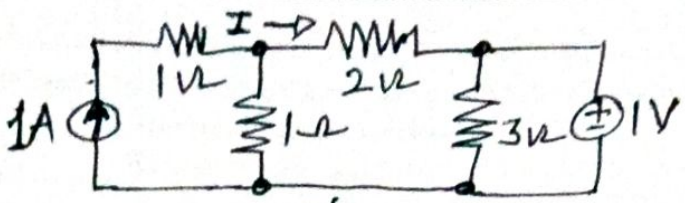
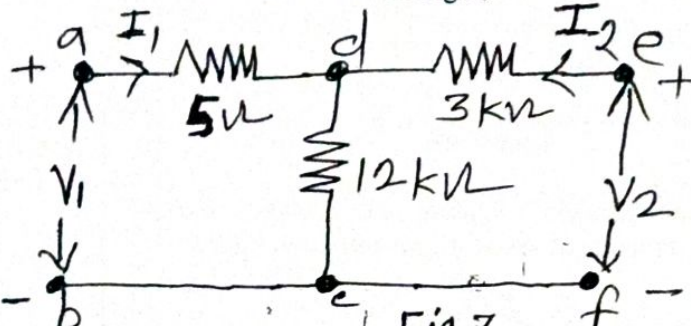
Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication Engineering
Summer Term Examination,
Academic Year: 2024-25

Class: S.Y.B. Tech	Subject: Network Theory	Subject Code: PCC-EC203
Date: 20/06/2025	Time: 14.30 to 17.30	Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- All questions are compulsory.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Questions	Marks	CO	BT
Q.1	State and explain Kirchhoff's current law. And also find the magnitude and direction of the unknown current in Fig.1.  Fig.1.	10	CO1	BT1
Q.2	What is resonance? What are the properties of a resonance RLC series circuit.	10	CO4	BT2
Q.3	A $10\ \mu\text{F}$ capacitor is initially charged to 100 V D.C. it is then discharged through a resistance of R ohms for 20 seconds when the p.d. across the capacitor is 50 V. Calculate the value of R .	10	CO3	BT4
Q.4	What are the key element correspondences in the Duality Principle for network theory? A series circuit with two resistors, $R_1=10\ \Omega$ and $R_2=20\ \Omega$, connected to a voltage source $V=10\ \text{V}$. Find the dual network and calculate the current in the dual circuit if the original network's total current is $I=2\ \text{A}$.	10	CO2	BT5
Q.5	Find the current in a series R-L circuit having $R=2\ \Omega$ and $L=10\ \text{H}$ while a d.c. voltage of 100V is applied. What is the value of this current after 5 seconds of switching on.	10	CO5	BT3
Q.6	State and explain Thevenin's theorem.	10	CO2	BT3
Q.7	Find I in the circuit shown in Fig.2 by using superposition theorem.	10	CO4	BT4

		 Fig. 2.			
Q.8		A $50\mu\text{F}$ capacitor, when connected in series with a coil having $40\ \Omega$ resistance, resonates at $1000\ \text{Hz}$. Find the inductance of the coil. Also obtain the circuit current if the applied voltage is 100V . Also calculate the voltage across the capacitor and the coil at resonance.	10	CO3	BT5
Q.9		Write a note on filter circuit	10	CO5	BT3
Q.10		Derive the expression for the Z parameter. Also find the Z parameters for the network shown in Fig. 3.  Fig. 3.	10	CO4	BT4

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO 1	Understand basics electrical circuits with nodal and mesh analysis.
CO 2	Appreciate electrical network theorems.
CO 3	Apply Laplace Transform for steady state and transient analysis.
CO 4	Determine different network functions.
CO 5	Appreciate the frequency domain techniques.